Online Account Opening

Collaboration Development

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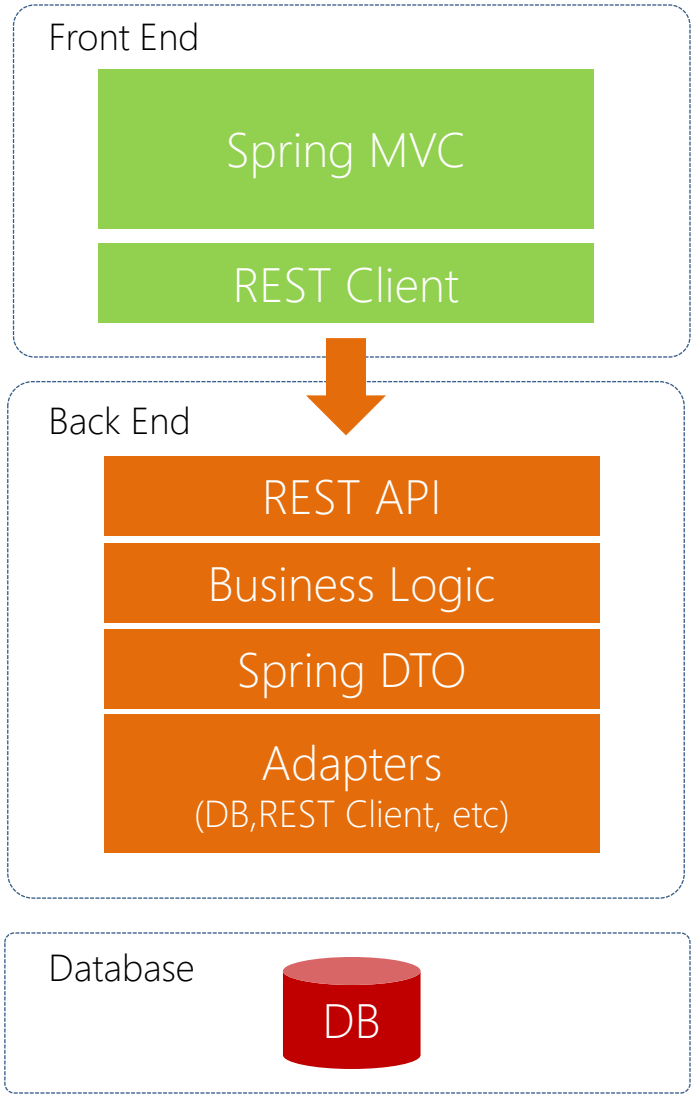
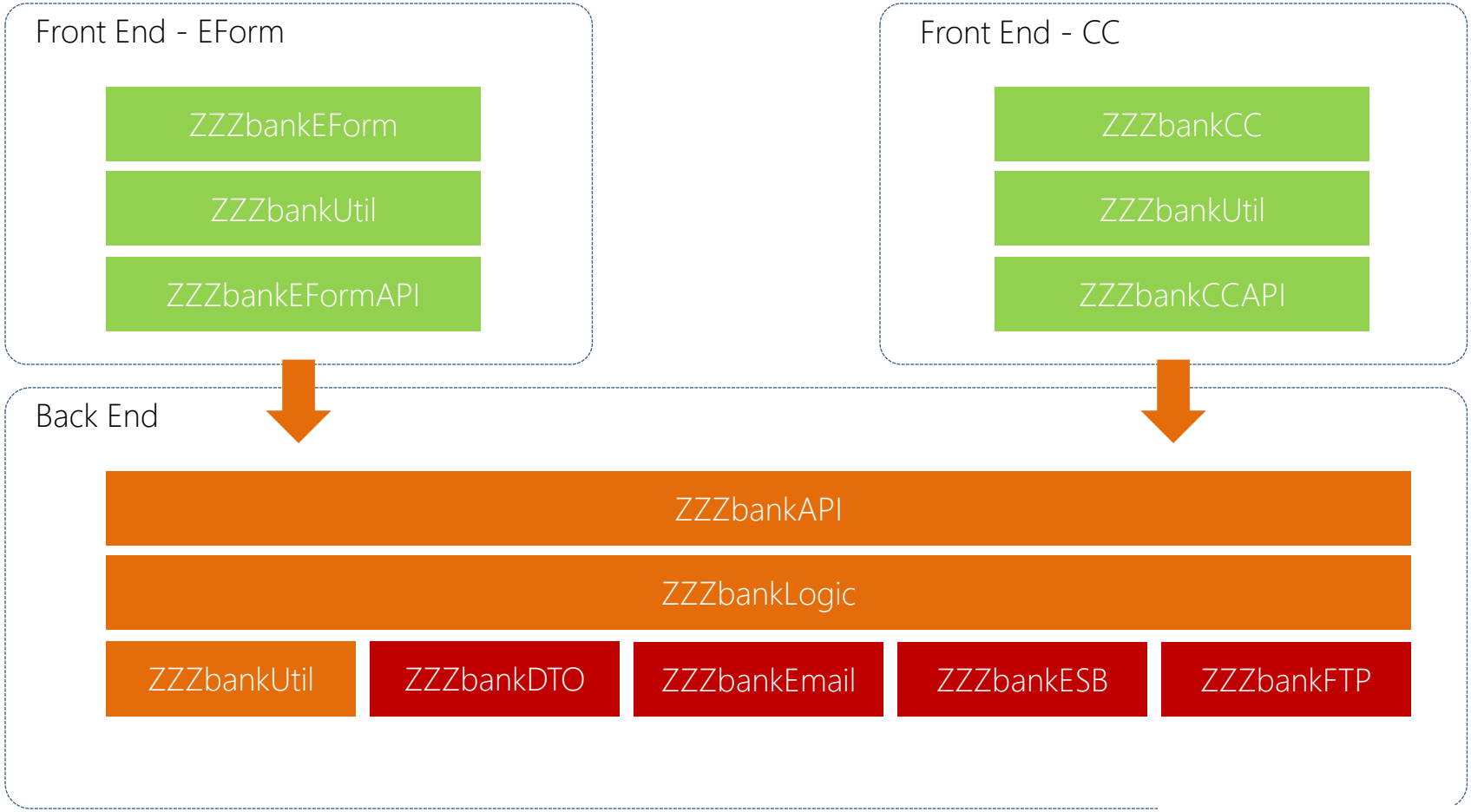
DEVELOPMENT CONVENTION

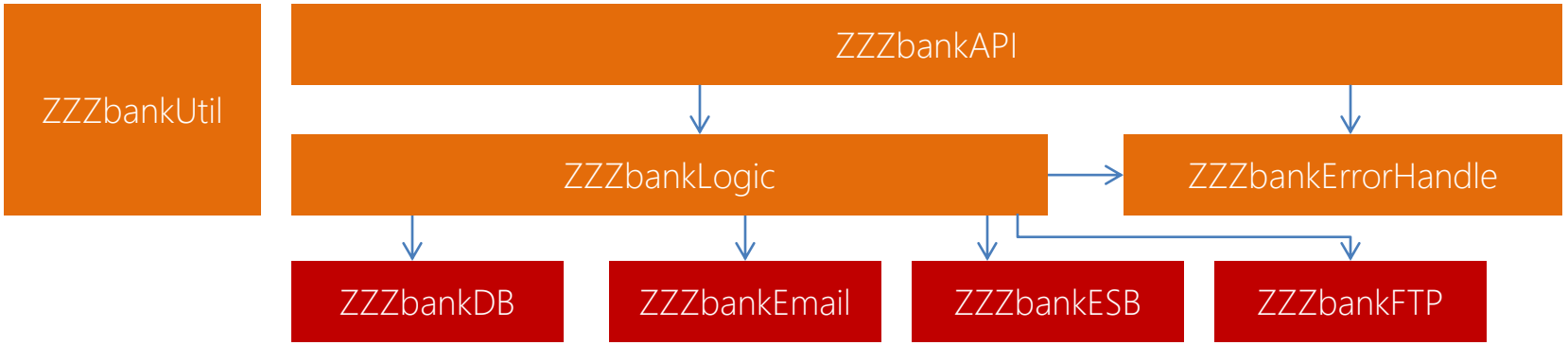
High Level Architecture Design



Module Architecture Design

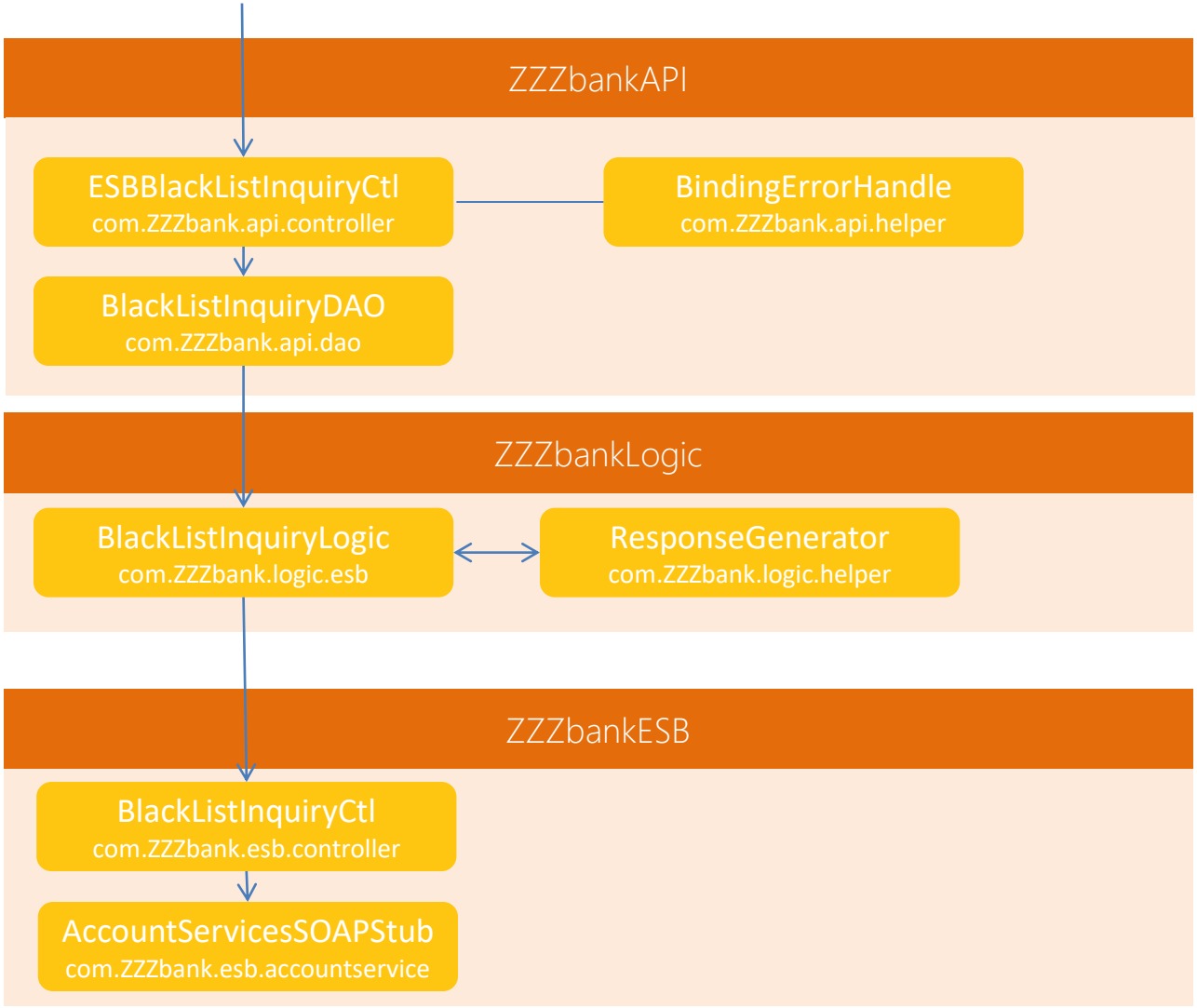


STP Backend Architecture



- ZZZbankUtil is used by any other component
- External System Access STP Backend thru ZZZbankAPI
- ZZZbankAPI accessing adapter through ZZZbankLogic
- All Error Message shall be transformed to proper Response

Message by ZZZbankErrorHandle


ZZZbankAPI
ESBBlackListInquiryCtl
com.ZZZbank.api.controller
BindingErrorHandle
com.ZZZbank.api.helper
BlackListInquiryDAO
com.ZZZbank.api.dao
ZZZbankLogic
BlackListInquiryLogic
com.ZZZbank.logic.esb
ResponseGenerator
com.ZZZbank.logic.helper
ZZZbankESB
BlackListInquiryCtl
com.ZZZbank.esb.controller
AccountServicesSOAPStub
com.ZZZbank.esb.accountservice

Error Handle Sample

ZZZbankLogic

```
// found error
respHeader =
respMessageController.getResponseHeader(null,"java.net.timeOutException","ESB")'
```

ZZZbankErrorHandle

```
com.ZZZbank.errorhandle.controller.ResponseMessageController

public Hashtable getResponseHeader(String code,String message, String module){
        Hashtable responseHeader = new Hashtable();
        if (code == null) || (code.equalsIgnoreCase("")){
                // get response by code
                String respDesc = getRespMessageByCode (module,code);
        } else {
                // get response by message
                String respDesc = getRespMessageByMessage (module,message);
        }
        return responseHeader;
}
```

Architecture Consideration

- 3 tier Application, seggregation between
  - Front-End
  - Back-End
  - and Data Layer (DB)
- Security Compliance
- Modularity
  - Code Classification
  - Ease On Future Development
  - Maintainability

Code Convention

- Following Java Convention Name, sample :
  - Project : refer to naming from SUN → ZZZbankDTO
  - Package : com.ZZZbank.eform
  - Class : BaseController
  - Method : getAccountInformation
  - Variable : accountNo
  - Constants : ACCOUNT_NO

Java Project Definition

- ZZZbankDTO : DTO Classes
- ZZZbankLogic : Application logic classes which call DTO or other Service Adapters to external system
- ZZZbankEForm : E-Form Classes and Web Artifact (Spring MVC)
- ZZZbankCC : Customer Care Classes and Web Artifact (Spring MVC)
- ZZZbankAPI: Backend Server API, conduct direct call to Logic Classes
- ZZZbankESB: ESB Integration Adapter Classes
- ZZZbankEmail: Email Integration Classes
- ZZZbankFTP: FTP Integration Classes
- ZZZbankUtil: Multi Purpose Utility Classes

RESTful Web Service Response

@HTTPHeader

HTTP 200 OK

@ResponseBody

```
{
"Response":
    {
      "RespHeader":
          {
               "respCode":"1",
                   "respDesc":"success"
          },
      "RespBody":
          {
                   "highRiskFlag":"true"
           }
    }
}
```

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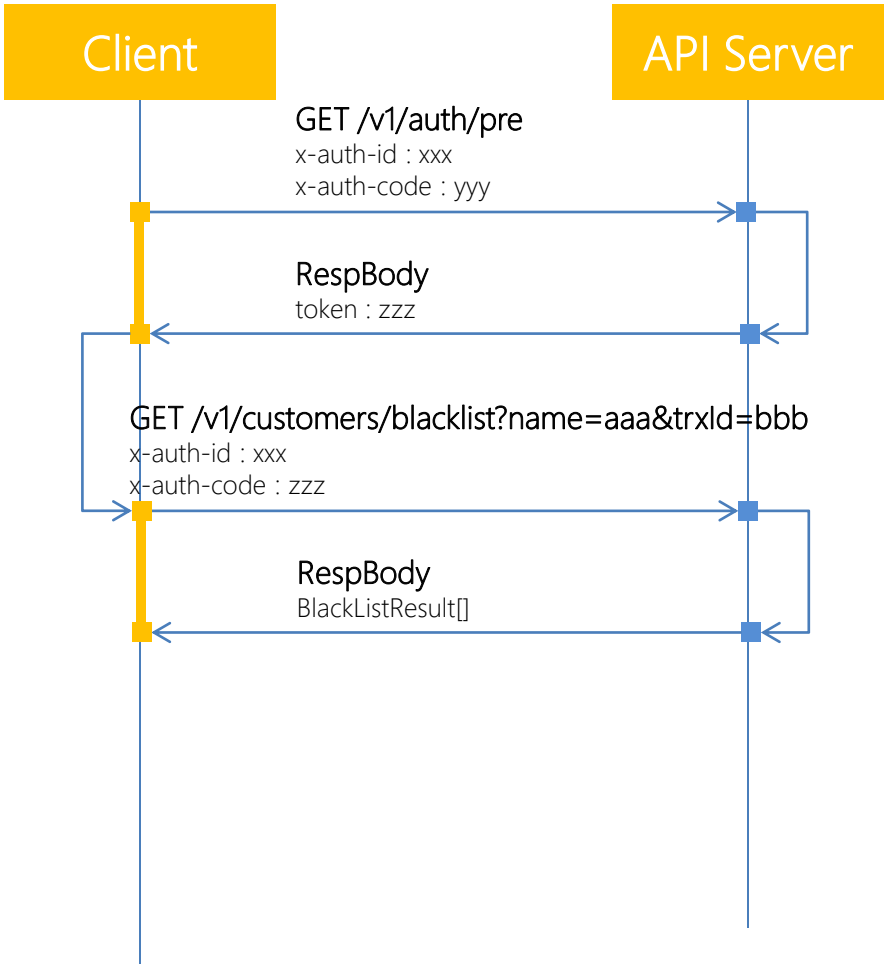
SECURED INTERFACE

REST API Security

- Leverage JWT (JSON Web Token) as security solution
- Benefits : does not store on Server, lesser space on Storage and Memory
- How does it works :
- API Call 1 :
  - Client call Pre-Authenticate API, provide secret key, combined with client datetime
  - API Server validate the secret key and provide JWT String should it be valid
- API Call 2 :
  - Client utilize JWT as one of header param for subsequent request
  - API Server validate the JWT Session
  - If JWT String Valid, Server shall call the "integration interface" and set respCode = 1
- Any failure in validation shall be responded with proper HTTP Status Code (e.g : 400 – Bad Request, 401 – Unauthorized)



Pre-Authenticate Code

com.ZZZbank.api.controller.AuthSecurityCtl

```
@RequestMapping(value="/auth/pre",produces="application/json",method=RequestMethod.GET)
@ResponseBody
private ResponseEntity getJwtCode(@RequestHeader Map<String,String> headers){
...
Hashtable htResult = authDAO.doValidatePreAuth(headers);
...
responseEntity = new ResponseEntity(htResult,HttpResponseCode.getHttpResonseCode(respCode));
...
}
```

REST WS

com.ZZZbank.api.dao.AuthSecurityDAO

```
public Hashtable doValidatePreAuth(Map<String,String> mapHeader){
                    return AuthHandler.doPreAuth(mapHeader);
}
```

DAO

com.ZZZbank.api.security.AuthHandler

```
public static Hashtable doPreAuth(Map<String,String> mapHeader){
...
if ((strCmp==strSecretKey)||(strCmp.equals(strSecretKey))){
                    String strJwtCode = JwtHandler.doGenerateJwt(strSecretKey, systemId, userId);
...
```

Security

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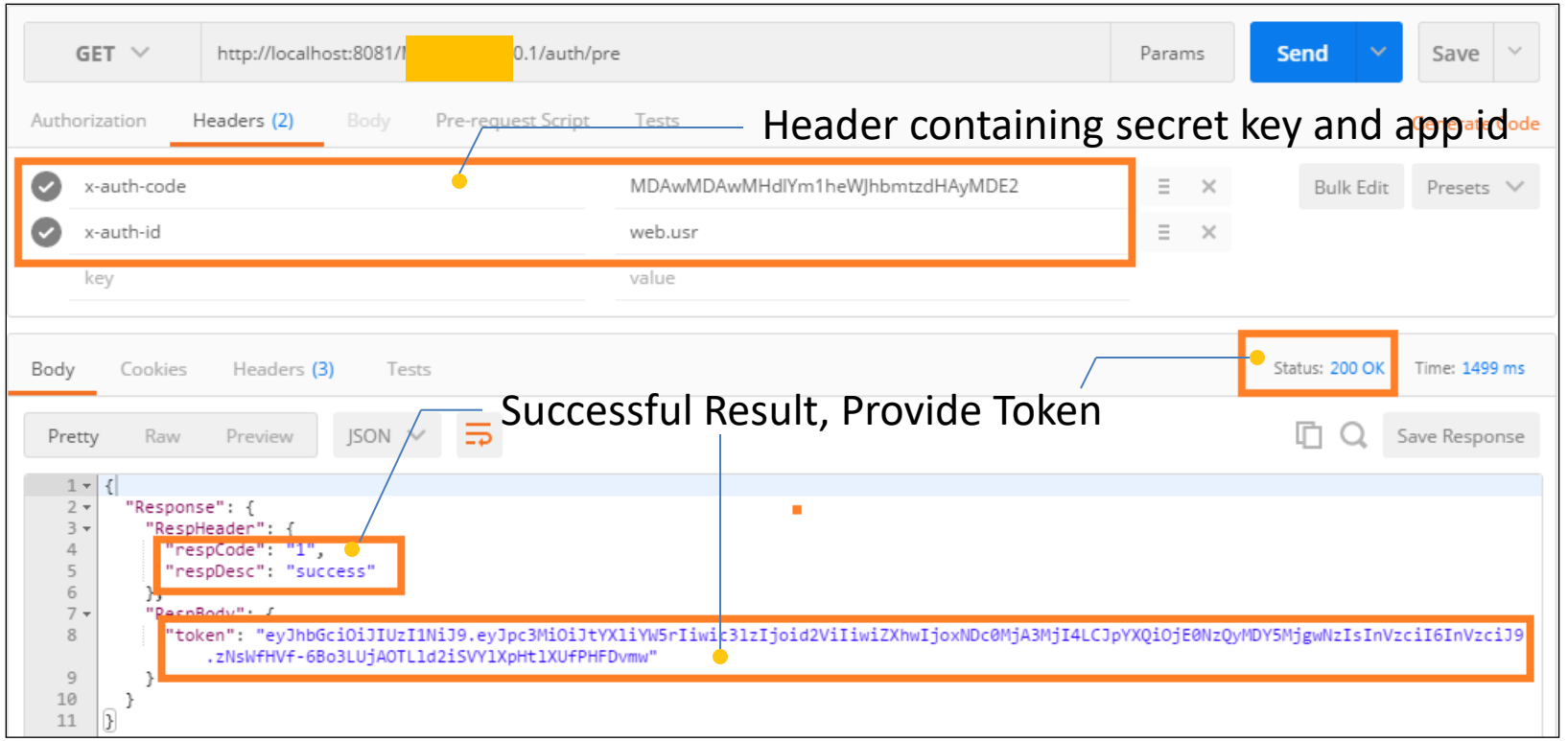
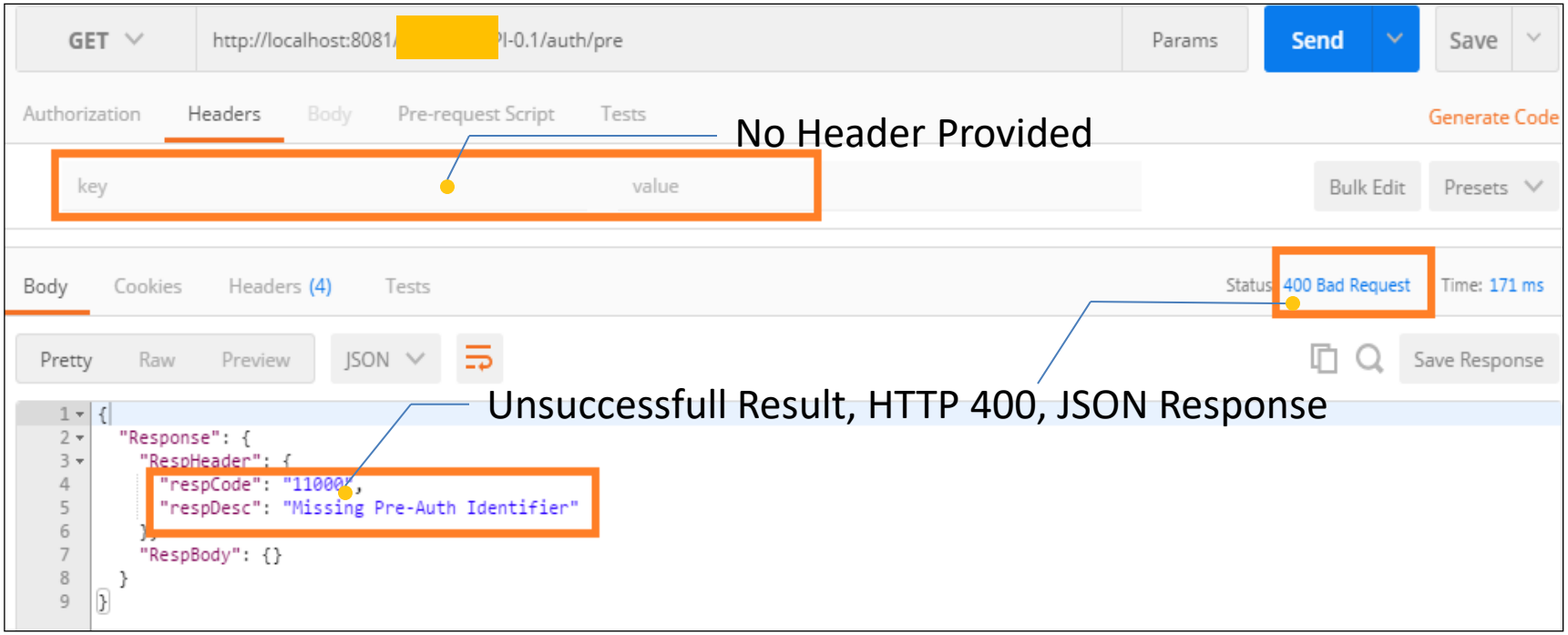
SAMPLE ESB INTEGRATION

Pre-Auth Sample : Positive Case



Pre-Auth Sample : Negative Case



BlackListInquiry (assumed Token Provided)

- Request :

http://localhost:8081/ZZZbankAPI-0.1/customers/blacklist?name=michael&trxID=001

- Response :

{"Response":{"RespHeader":{"respCode":"1","respDesc":"success"},"RespBody":{"blackListCustomers":[{"id":"201600001","currentRecordVersion":null,"taxNo":null,"name":"MICHAEL JACKSON","birthDate":"1972-08-30","sourceCode":null,"remark":null,"registrationNo":"2003012300001","addrLine1":null,"addrLine2":null,"addrLine3":null,"blacklistCode":null,"blacklistNo":null,"letterNo":null,"letterDate":null,"createdBy":null,"createdDate":null,"updatedBy":null,"updatedDate":null,"maintenanceBranchCode":null,"operatorId":null,"telephoneNo":null,"historyLetterNo":null,"historyDateLetter":null,"historyName":null,"historyAddrLine1":null,"historyAddrLine2":null,"historyAddrLine3":null,"mtRsnCd":null,"mtBlacklistStatus":null,"motherMaidenName":null,"acctNo":null,"status":null,"clearanceDate":null,"asAtDate":null,"downloadDate":null,"hostSourceMappingList":null,"sourceDescription":null,"expiryDate":null,"nationalityCode":null,"nationalityDescription":null,"companyName":null,"customerName":null,"cifNumber":null,"systemFlag":null,"terroristBirthDate":null,"blacklistFlag":null,"mothersMaidenNameCheck":null,"birthDateCheck":null,"regNoCheck":null},{"id":null,"currentRecordVersion":null,"taxNo":null,"name":"MICHAEL LEARNS TO ROCK","birthDate":"1975-12-31","sourceCode":null,"remark":null,"registrationNo":"20000201678","addrLine1":null,"addrLine2":null,"addrLine3":null,"blacklistCode":null,"blacklistNo":null,"letterNo":null,"letterDate":null,"createdBy":null,"createdDate":null,"updatedBy":null,"updatedDate":null,"maintenanceBranchCode":null,"operatorId":null,"telephoneNo":null,"historyLetterNo":null,"historyDateLetter":null,"historyName":null,"historyAddrLine1":null,"historyAddrLine2":null,"historyAddrLine3":null,"mtRsnCd":null,"mtBlacklistStatus":null,"motherMaidenName":null,"acctNo":null,"status":null,"clearanceDate":null,"asAtDate":null,"downloadDate":null,"hostSourceMappingList":null,"sourceDescription":null,"expiryDate":null,"nationalityCode":null,"nationalityDescription":null,"companyName":null,"customerName":null,"cifNumber":null,"systemFlag":null,"terroristBirthDate":null,"blacklistFlag":null,"mothersMaidenNameCheck":null,"birthDateCheck":null,"regNoCheck":null},{"id":null,"currentRecordVersion":null,"taxNo":null,"name":"MICHAEL JORDAN","birthDate":"1968-01-23","sourceCode":null,"remark":null,"registrationNo":"19980100001","addrLine1":null,"addrLine2":null,"addrLine3":null,"blacklistCode":null,"blacklistNo":null,"letterNo":null,"letterDate":null,"createdBy":null,"createdDate":null,"updatedBy":null,"updatedDate":null,"maintenanceBranchCode":null,"operatorId":null,"telephoneNo":null,"historyLetterNo":null,"historyDateLetter":null,"historyName":null,"historyAddrLine1":null,"historyAddrLine2":null,"historyAddrLine3":null,"mtRsnCd":null,"mtBlacklistStatus":null,"motherMaidenName":null,"acctNo":null,"status":null,"clearanceDate":null,"asAtDate":null,"downloadDate":null,"hostSourceMappingList":null,"sourceDescription":null,"expiryDate":null,"nationalityCode":null,"nationalityDescription":null,"companyName":null,"customerName":null,"cifNumber":null,"systemFlag":null,"terroristBirthDate":null,"blacklistFlag":null,"mothersMaidenNameCheck":null,"birthDateCheck":null,"regNoCheck":null}],"totalPages":1}}}

CustomerHighRiskInquiry (assumed Token Provided)

- Request :

http://localhost:8081/ZZZbankAPI-0.1/customers/highrisk?name=michaels&trxID=001&occupationCode=AD&ecoSectorCode=AD

- Response :

{"Response":{"RespHeader":{"respCode":"1","respDesc":"success"},"RespBody":{"highRiskFlag":"true"}}}